

Handout: *Loux & Crisp, “Causation”*

The Problem: What Is a Cause?

The central metaphysical dispute framing this chapter is whether causation is an irreducibly modal relation—a relation of *necessary connection*—or whether it can be analyzed in nonmodal terms.

- Traditional metaphysicians: Causes involve *power*, *force*, or *necessitation*.
- Hume: No impression of necessary connection → causation reduces to constant conjunction.
- Neo-Humeans: Seek refined nonmodal analyses.
- Mackie’s project: Preserve Hume’s nonmodal spirit while fixing the deficiencies of simple regularity theory.

The key question guiding this handout:

Can we explain ordinary causal judgments without appealing to mysterious modal necessity—yet avoid the counterexamples that plague Hume’s original account?

I. From Hume to Neo-Humeanism

1. Hume’s Reduction

Hume’s empiricist principle:

- Every idea must derive from an impression.
- No impression of necessary connection is found in sensation (billiard balls) or reflection (volition).
- Therefore, “necessary connection” is unintelligible.

Positive proposal:

- Causation = constant conjunction.

- Events of type K_1 are regularly followed by events of type K_2 .

But Hume's view faces two main objections (pp. 186–87):

(a) Too Broad

Regular sequences that are not causal (day/night; factory horn).

(b) Singular Causation

Historical and one-off causal claims resist generalization.

The inadequacy of crude regularity motivates refinement.

II. Mackie's INUS Theory: Structured Regularity

1. The Core Motivation

Mackie agrees with Hume that:

- We should not posit mysterious causal powers.
- Causation should be analyzable in nonmodal terms.

However, Hume's constant conjunction is too crude. It cannot distinguish:

- Genuine causal laws from accidental regularities.
- Foreground causes from background conditions.

Mackie's solution: analyze causation in terms of necessary and sufficient conditions, but relativized to a causal field (pp. 190–92).

2. The Causal Field

A causal claim answers a *why-question*.

But such questions are always incomplete.

Example:

“Why did the house burn down?”

Possible causal fields:

- The building's history.
- Competing possible ignition sources.
- Comparison with similar houses.

The causal field fixes:

- What counts as background.
- What contrasts are relevant.

Thus, causes are not metaphysically absolute; they are contextually identified difference-makers.

3. INUS Conditions Defined

Mackie's famous formula:

A cause is an Insufficient but Necessary part of an Unnecessary but Sufficient condition (p. 191).

Let's analyze this step by step.

(A) Insufficient

The short circuit alone did not cause the fire.

(B) Necessary part

Within a specific bundle:

(short circuit + oxygen + dry rags + no sprinklers)
the short circuit was indispensable.

(C) Unnecessary bundle

Other bundles could have caused the fire.

(D) Sufficient bundle

The actual bundle present guaranteed the fire.

So:

- The short circuit is not a metaphysical necessitator.
 - It is a structural component in a complex sufficient set.
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4. Logical Form

Mackie analyzes necessary and sufficient conditions via conditionals (pp. 191–92).

Necessary:

If x had not occurred, y would not have occurred.

Sufficient:

Since x occurred, y occurred.

He treats these as telescoped arguments backed by general regularities.

Thus, Mackie attempts to remain within the Humean framework:

- No irreducible modal glue.
 - Just structured regularities plus conditional reasoning.
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5. Strengths of Mackie

✓ Captures Causal Complexity

Real-world causation is conjunctive and multi-factorial.

✓ Explains Selectivity

Why we single out one cause:

- It is the relevant difference-maker within the field.

✓ Improves on Hume

Not mere constant conjunction, but structured regularity.

6. Lewis's Objections to Mackie

Loux & Crisp emphasize two problems (pp. 192–93).

(A) Epiphenomenal Effects

Structure:

$a \rightarrow b$

$a \rightarrow c \rightarrow d$

b has no effects. Yet on INUS:

- b may qualify as a cause of d.

Problem:

Causally inert events become causes.

(B) Preemption

Two potential causes:

a and b.

a preempts b.

Only a actually causes c.

But b may still satisfy INUS conditions.

Problem:

Preempted causes are misclassified.

These cases motivate Lewis's departure from regularity.

III. Lewis's Counterfactual Theory

Lewis shares Mackie's ambition:

- Provide a nonmodal account.
- Avoid mysterious necessitation.

But he rejects the regularity framework.

1. Core Idea

Lewis develops a suggestion found in Hume's Enquiry:

If the cause had not been, the effect would not have existed (p. 193).

Thus:

c causes e iff: If c had not occurred, e would not have occurred.

Causation = counterfactual dependence.

2. Counterfactual Semantics

To analyze counterfactuals, Lewis introduces:

- Possible worlds (but not in the Lowe sense!)
- Ordered by comparative similarity to the actual world (pp. 194–95).

A counterfactual:

“If p were the case, q would be the case”

is true iff:

- In the closest p-world(s), q is also true.

Applied to causation:

c causes e iff

in the closest world where c does not occur, e does not occur either.

3. Causal Chains

Lewis defines:

- e depends on c $\rightarrow$ counterfactual holds.
- c causes e $\rightarrow$ there exists a chain of counterfactual dependence linking them (p. 195).

This handles:

- Indirect causation.
 - Multi-step processes.
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4. Lewis's Response to Mackie-Type Problems

(A) Epiphenomena

Lewis rejects backtracking counterfactuals:

- The past does not counterfactually depend on the future.

Thus:

If b had not occurred, a would not have occurred
is illegitimate.

Once backtracking is prohibited:

- Epiphenomenal events are not causes.
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(B) Preemption

Similarly:

Preempted causes fail to satisfy counterfactual dependence.

Lewis blocks misclassification by careful constraints on similarity ordering and non-backtracking.

5. Remaining Problem: Overdetermination

Two shooters kill simultaneously.

If either had not fired:

- The victim still would have died.

Thus:

No counterfactual dependence.

Lewis:

- Our intuitions here are unclear.
 - These are not decisive counterexamples (p. 196).
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The Role of Modality in the Hume–Mackie–Lewis Debate

Thinker	View of Modality	How Causation Is Analyzed	Does the View Require Irreducible Modality?	What's at Stake
Hume	Suspicious of modality; no impression of necessary connection	Constant conjunction + psychological projection	No — modality is rejected as unintelligible	If modality is irreducible, Hume's reduction fails
Mackie	Tries to avoid primitive necessity	INUS conditions (insufficient but necessary part of an unnecessary but sufficient condition)	Unclear / Debated — depends on whether "necessary" and counterfactuals are reducible	If counterfactuals are irreducibly modal, Mackie hasn't really eliminated modality
Lewis	Modality is reducible	Counterfactual dependence via similarity-ordered possible worlds	No (officially) — modality reduced to quantification over concrete worlds	If possible worlds analysis fails, modality may be irreducible after all
Anti-Humean (Kantian / Aristotelian)	Modality is fundamental	Causation involves real necessary connection	Yes — necessity is primitive	If modality is irreducible, this view gains support

The Fork in the Road

If Modality Is...

Then...

Irreducible

Causation likely involves genuine necessary connections. Humean reductionism fails.

Reducible

Causation can be analyzed nonmodally (via regularities or possible worlds). Humean ambitions succeed.